MAF PORTFOLIO · 2026

Muhamad Agus Firdaus

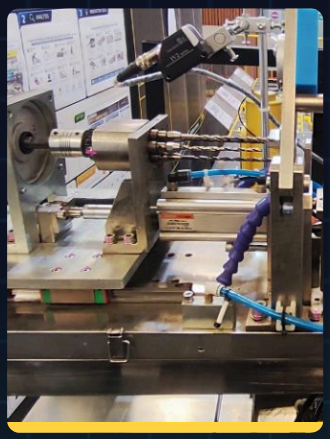
Mechanical & Electrical Engineer

- PLC Programmer
- Automation System Designer
- SolidWorks Mechanical Designer

Designing control systems from scratch — from PLC logic and panel wiring to production-ready mechanical design.



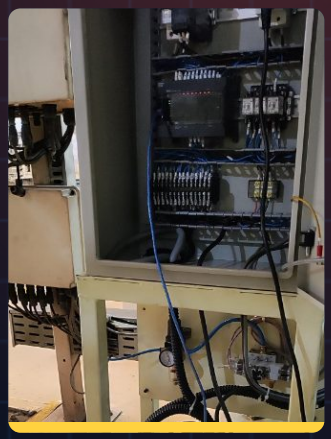
ELECTRICAL
Torque Check Machine Control System



ELECTRICAL
Camshaft Deburring (Baritori) Machine Control System



ELECTRICAL
OK / NG Check with Camera



ELECTRICAL
Automatic Parts Rack Cover

1+
Years in industrial automation

11
Automation & mechanical projects delivered

2
PLC platforms: Omron & Mitsubishi

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Bekasi, West Java, Indonesia

About me

I'm a Mechanical & Electrical Engineer with more than a year of experience in industrial automation: control panel assembly, machine wiring, PLC programming and modification (Omron & Mitsubishi), and collaborative robot setup.

I regularly work on custom machine projects for clients in the manufacturing industry. A Mechatronics Engineering graduate looking to keep growing as an Automation Engineer.

LOCATION	EDUCATION	STATUS
Bekasi, West Java	Mechatronics Engineering — SMK YPPT Garut	Open to projects & roles

Skills

Advanced · Intermediate · Basic

PLC & HMI

- Omron · CX-Programmer — Advanced
- Omron · CX-Designer — Advanced
- Mitsubishi · GX Works — Advanced

Electrical

- Control panel assembly — Advanced
- Machine wiring — Advanced
- See Electrical — Advanced

Mechanical, CAD & Robotics

- SolidWorks — Advanced
- AutoCAD — Advanced
- Pneumatics — Advanced
- Cobot Rokae xMate CR12 — Advanced

Soft Skills & Other

- Disciplined — Advanced
- Detail-oriented — Advanced
- Analytical — Advanced
- Time management — Advanced
- Microsoft Office — Advanced

Project index

Electrical · Mechanical

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ELECTRICAL

Torque Check Machine Control System

Role: PLC Programmer & Panel Builder

Converted a previously manual torque check machine to automatic operation for a manufacturing client.

THE PROBLEM

The torque check process was previously run manually.

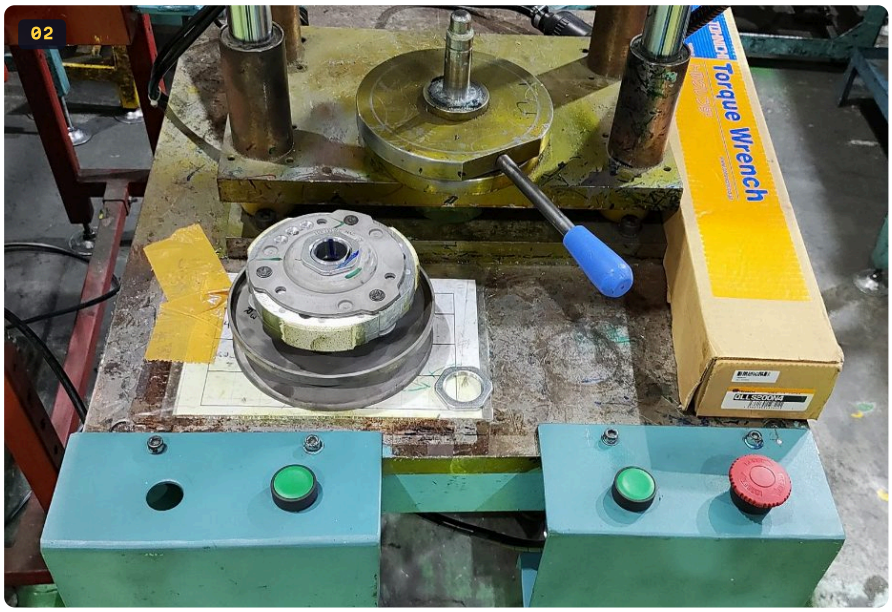
WHAT I DID

- Selected the control system components
- Assembled the control panel
- Wrote the PLC program

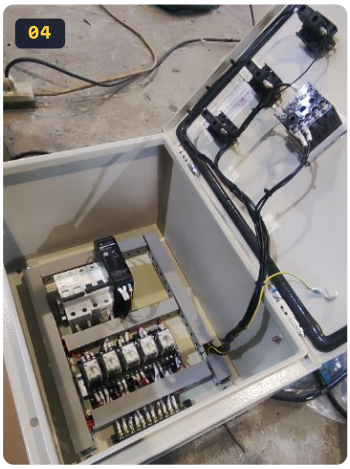
TOOLS

PLC | Control Panel | Counter Omron H7CC













01 The torque check machine with its new control panel: three status lamps, an Omron H7CC counter and a buzzer.

02 Work table: the part on its jig, two green push buttons, an emergency stop and the torque wrench box.

03 Indicator panel: green, yellow and red lamps, an Omron H7CC counter and a buzzer.

04 Inside the panel during assembly: MCBs, power supply, five relays and terminals.

05 Installation in progress: running the wiring to the button boxes at the front of the machine.

06 One full cycle after installation: the part is loaded on the jig, processed, then its torque is checked with the torque wrench wired to the panel. (video, watch it on the website)

ELECTRICAL

Camshaft Deburring (Baritori) Machine Control System

Role: PLC Programmer & Panel Builder

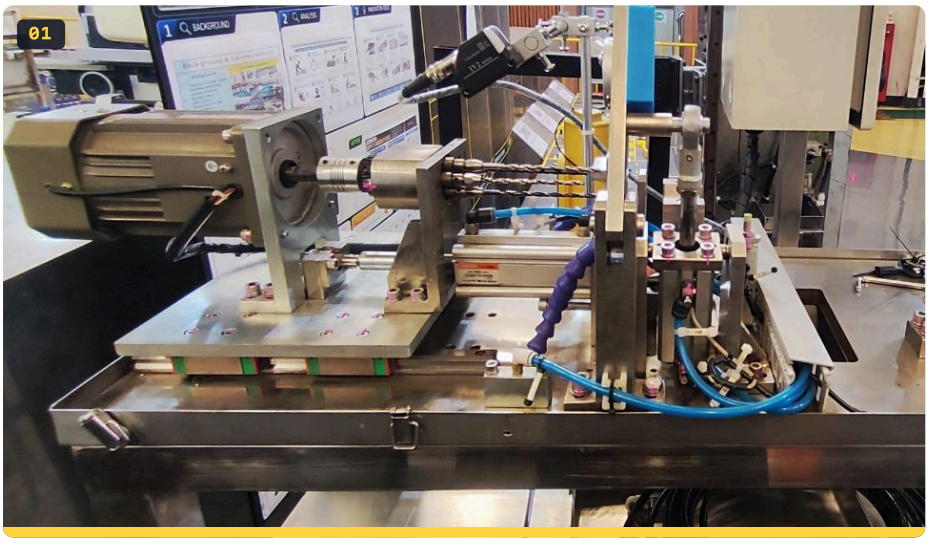
Control system for a camshaft deburring (baritori) machine for a manufacturing client: Omron CP2E PLC, operator panel and a vision sensor that checks the drill bits.

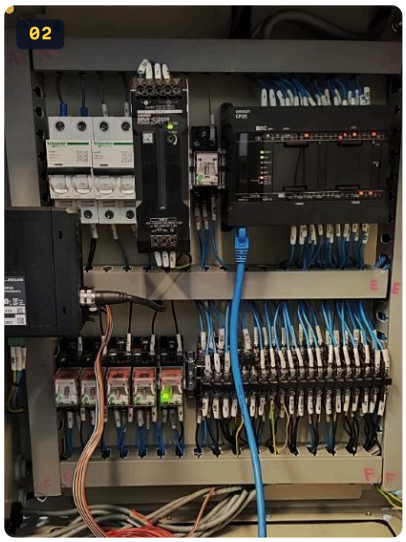
WHAT I DID

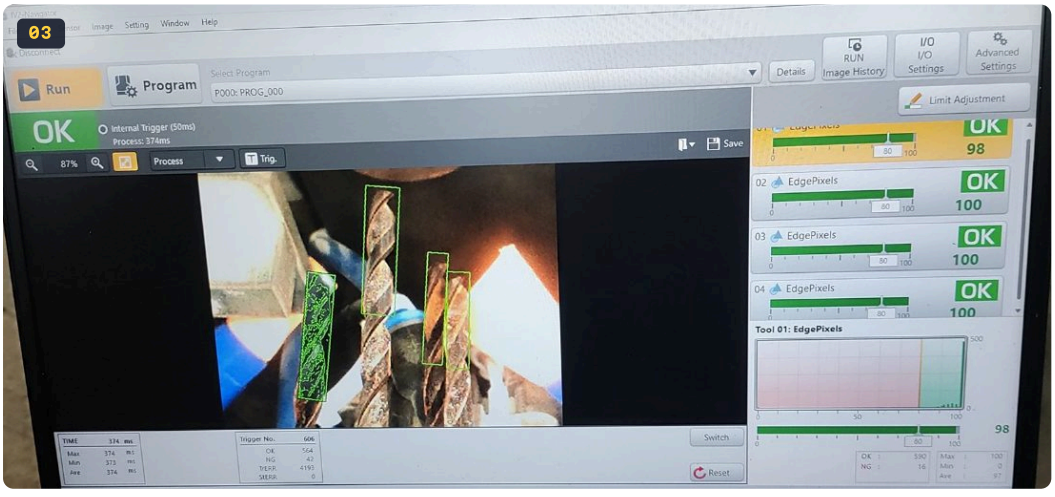
- Selected the control system components
- Assembled the control panel: Omron CP2E PLC, power supply, relays and labelled terminals
- Wrote the PLC program: manual/auto modes, home position and alarms
- Set up the Keyence IV2 vision sensor to check the drill bits

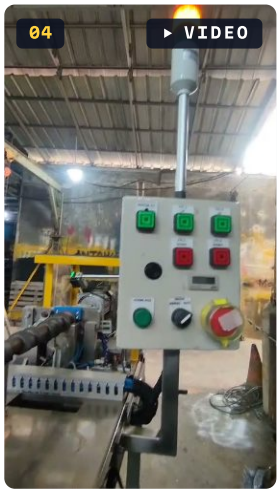
TOOLS

PLC Omron CP2E | Control Panel | Vision Sensor Keyence IV2









01 The deburring unit: a motor driving four drill bits, pneumatic cylinders, and a vision sensor watching the bits from above.

02 Inside the control panel: Omron CP2E PLC, Omron S8VK 24 VDC power supply, MCBs, relays and terminals with labelled wiring.

03 Keyence IV2 vision sensor setup: four inspection windows on the drill bits, all reading OK.

04 Workshop test run: the operator panel with motor and cylinder lamps, manual/auto mode, home position button and tower lamp. A camshaft is loaded on the V-blocks. (video, watch it on the website)

ELECTRICAL

OK / NG Check with Camera

Role: Panel Builder & Wiring

An OK/NG indicator box for balancing machines: a camera reads the result on the machine's screen, and lamps and counters show whether the part is OK or NG.

WHAT I DID

- Built the indicator box: power lamp, OK and NG lamps, two Omron H7EC counters and a buzzer
- Mounted the camera facing the balancing machine's result screen
- Wired the camera output to the OK/NG lamps and counters
- Installed the same unit on balancing machines no. 1 and no. 2

TOOLS

Vision Sensor · Control Panel · Counter Omron H7EC









01 The indicator box installed on balancing machine no. 2: power lamp, OK and NG lamps each with its own counter, and a buzzer.

02 The camera aimed at the balancing machine's result screen.

03 The same unit on balancing machine no. 1.

04 Test run: the OK and NG lamps light in turn, followed by the balancing machine screen the camera reads. (video, watch it on the website)

ELECTRICAL

Automatic Parts Rack Cover

Role: Automation System Designer

A parts rack cover that only opens for the detected engine type, so operators can't pick the wrong part.

THE PROBLEM

Operators risked picking the part for the wrong engine type (A or B).

WHAT I DID

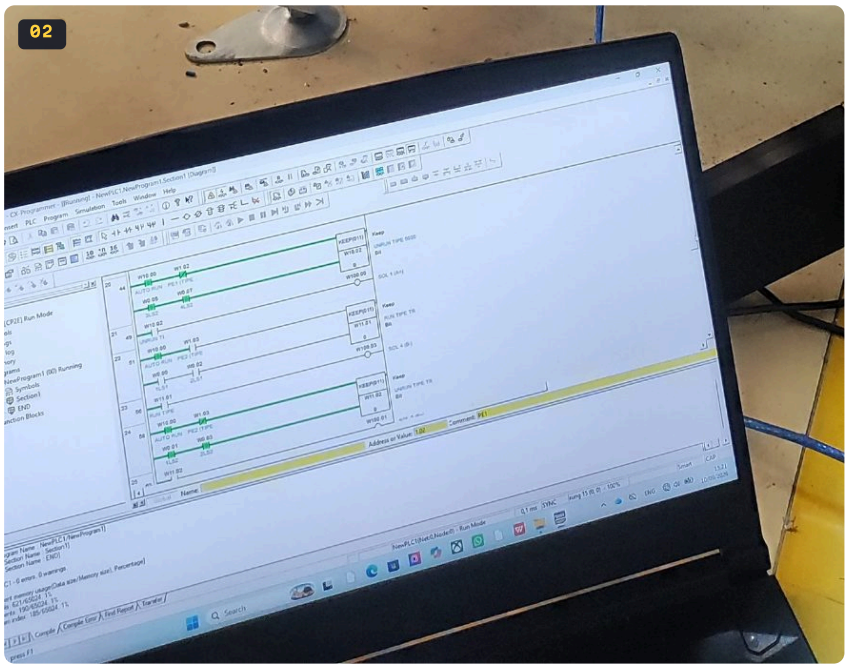
- Built an automatic cover on the parts rack
- Connected the cover to the existing machine sensor
- The cover only opens for the engine type that is read (A or B)
- Programmed the Omron CP2E PLC and checked the sequence online in CX-Programmer

TOOLS

PLC Omron CP2E · CX-Programmer · Pneumatics · Poka-Yoke



01 The rack cover's control panel: Omron CP2E PLC, relays and terminals. Below it, the solenoid valves that actuate the covers.



02 The ladder program monitored online in CX-Programmer during commissioning.

Bearing Press Machine Modification

Role: PLC Programmer & Wiring

Modified the Mitsubishi PLC program and panel wiring on a manufacturing client's 5-cylinder bearing press machine.

THE PROBLEM

The client's bearing press machine needed to work differently. The change had to be made on the PLC and panel already installed, without replacing the machine.

WHAT I DID

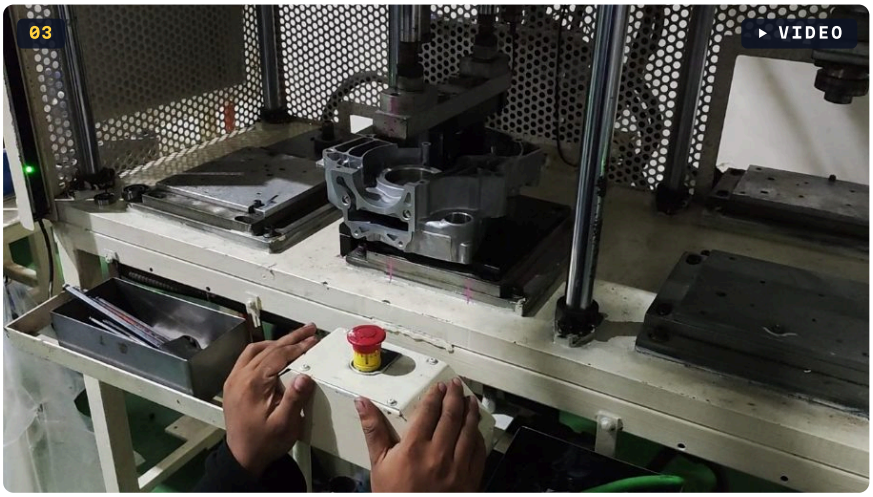
- Studied the machine's existing program and wiring before changing anything
- Modified the Mitsubishi FX3G-60M PLC program in GX Works
- Reworked the panel wiring: PLC I/O, relays and the operator panel
- Tested the press cycle on the machine until the sequence ran correctly

TOOLS

PLC Mitsubishi FX3G | GX Works | Wiring







01 Operator panel: status lamps, manual/auto and left/right mode selectors, and up/down controls for 5 cylinders.

02 Inside the control panel: Mitsubishi FX3G-60M PLC, relay bank, contactor and terminals.

03 Test run after the modification: the press cycle is started with both hands from the button box at the front of the machine. (video, watch it on the website)

ELECTRICAL

Rokae xMate CR12 Cobot Setup

Role: Setup & Troubleshooting

Set up a Rokae xMate CR12 cobot cell with a pneumatic gripper, sensors and roller conveyors, from I/O testing in the workshop to installation on site.

WHAT I DID

- Tested sensor inputs and solenoid valve outputs from the teach pendant
- Tested the cobot's motion with the pneumatic gripper
- Tested the roller conveyors before installation
- Installed and re-tested the whole cell on site

TOOLS

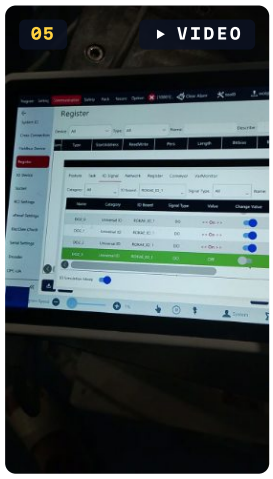
Cobot Rokae xMate CR12 · Pneumatics · Sensors · Conveyor

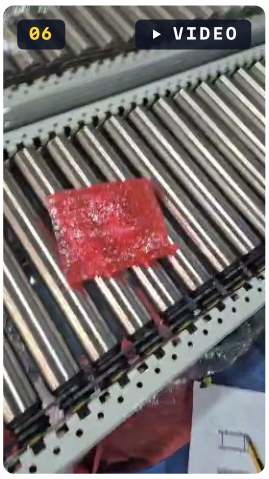














01 The cobot cell after installation: Rokae cobot, two station tables, teach pendant and roller conveyor.

02 Installation on site: the cobot on its base, with roller conveyors in front of and behind it.

03 Motion test in the workshop: the cobot and its pneumatic gripper move toward a test object on the station table. (video, watch it on the website)

04 Input test: a test object is placed in front of the station sensor while its status is watched on the teach pendant. (video, watch it on the website)

05 Output test: DO signals are switched on from the teach pendant and the solenoid valve LEDs light up. (video, watch it on the website)

06 Roller conveyor test in the workshop. (video, watch it on the website)

07 Test run after installation on site: a carton travels along the roller conveyor beside the cobot. (video, watch it on the website)

ELECTRICAL

Conveyor Repair

Role: Troubleshooting & Repair

Repaired a belt conveyor on a client's production line and got it running normally again, including checks on its control panel (Mitsubishi FX5U PLC and inverter).

WHAT I DID

- Checked the conveyor's control panel: Mitsubishi FX5U PLC, inverter and relays
- Repaired the conveyor until the belt ran again
- Test-ran the belt with a test object after the repair

TOOLS

PLC Mitsubishi FX5U | Inverter | Troubleshooting







01 The conveyor's control panel: Mitsubishi FX5U-32M PLC, inverter, relays and power supply.

02 Belt test run: checked at the end roller, then run with a test object on it. (video, watch it on the website)

03 The belt running again on the production line. (video, watch it on the website)

ELECTRICAL

Conveyor Modification

Role: Electrical

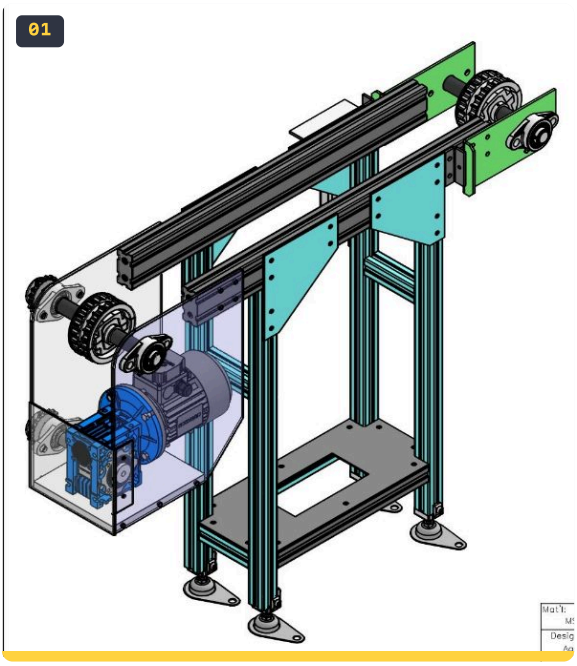
Modified a pallet conveyor on a client's production line: the unit was designed in 3D first, then installed and test-run on site.

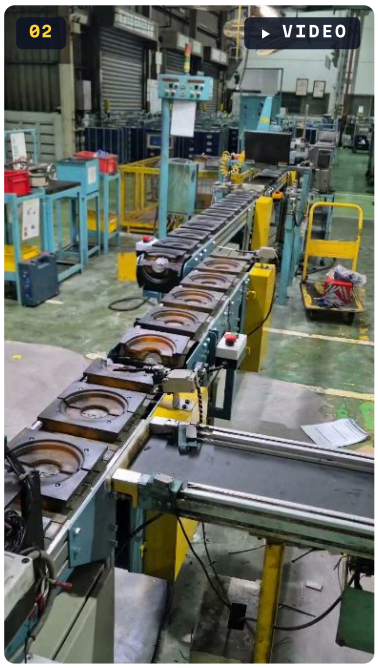
WHAT I DID

- Designed the conveyor unit in 3D: aluminium profile frame, geared motor, sprockets and bearings
- Installed the modification on the production line
- Test-ran the conveyor with pallets

TOOLS

SolidWorks · Mechanical Design · Conveyor







01 3D design of the conveyor unit: aluminium profile frame, geared motor, sprockets and bearings.

02 The pallet conveyor on the production line, seen from end to end. (video, watch it on the website)

03 Pallets turning over at the end of the conveyor while the line is running. (video, watch it on the website)

MECHANICAL

Machine Safety Fence

Role: Mechanical

Built and installed safety fencing around a press machine and its coil line at a client's plant.

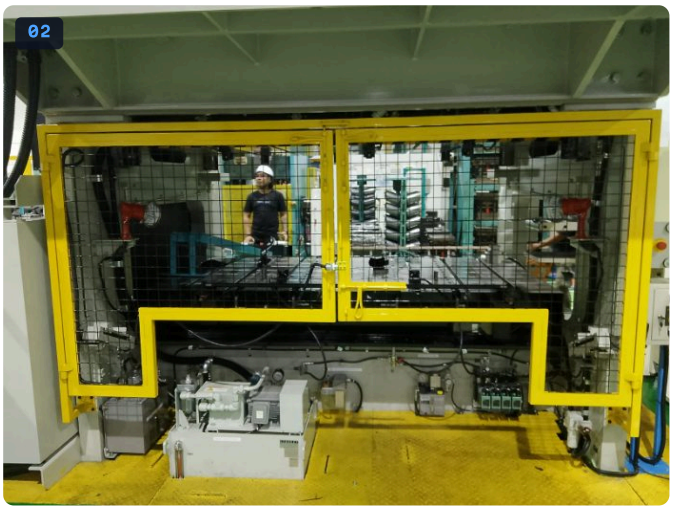
WHAT I DID

- Built steel-frame safety fencing with wire mesh for the press and coil line area
- Made access doors with latches where entry is needed
- Installed the fence on site and anchored it to the floor

TOOLS

Safety Fence











01 Gated fence at the uncoiler: the coil stays visible through the wire mesh.

02 Double doors at the rear of the press. The frame is shaped around the units beneath it.

03 Gated fencing around the coil line unit, next to the press.

04 Guard rail around the cabinet and cooling unit.

05 Side view: the fence at the uncoiler area, with the press behind it.

MECHANICAL

Safety Fence for 6 Production Lanes

Role: Mechanical

Built safety fencing for 6 production lanes at a client's plant.

WHAT I DID

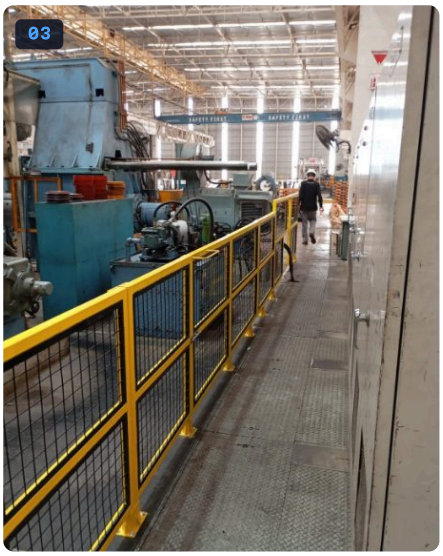
- Built the safety fence for 6 production lanes
- Made access doors with latches and handles
- Installed the fencing around the machines in each lane

TOOLS

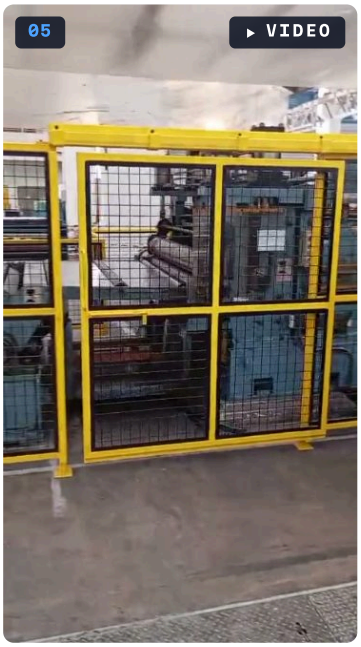
Safety Fence

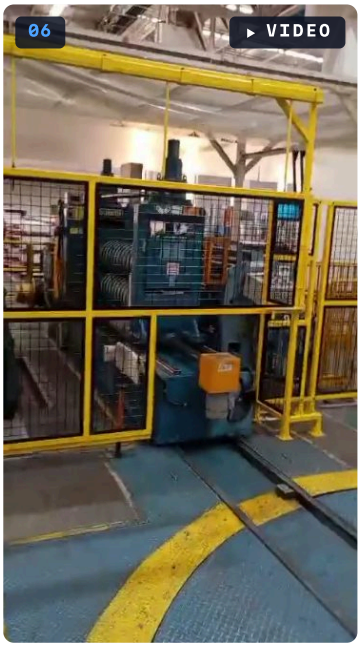












01 Fencing on one of the lanes: yellow frame with black wire mesh around the machine area.

02 Fencing along the slitter machine.

03 Fencing with a door along the walkway.

04 Fencing around the coil area of one lane.

05 Walkaround of the installed fencing: doors, latches and handles are tried one by one. (video, watch it on the website)

06 Walkaround of another lane: fencing and access doors around the machines. (video, watch it on the website)

Other projects

ELECTRICAL

PROJECT 11 / 11

Plate Cutting / Measuring Machine Control System

Role: PLC Programmer & Panel Builder

Control system for a plate cutting / measuring machine for a manufacturing client.

WHAT I DID

- Selected the control system components
- Assembled the control panel
- Wrote the PLC program

TOOLS

PLC · Control Panel

Experience & education

Sep 2025 — Present

Work

Mechanical & Electrical Engineer

PT Shimatek Shoji Indonesia

- Built the control system that converted a manual torque check machine to automatic (component selection, panel assembly, PLC programming)
- Built the control system for a camshaft deburring (baritori) machine (component selection, panel assembly, PLC programming)
- Built the control system for a plate cutting / measuring machine (component selection, panel assembly, PLC programming)
- Built an automatic parts rack cover linked to the existing machine sensor: it only opens for the engine type that is read (A or B), so operators can't pick the wrong part
- Modified customer machine programs and wiring using Mitsubishi PLCs (GX Works)
- Set up and troubleshot the Rokae xMate CR12 cobot
- Built a shutter
- Built safety fencing for 6 production lanes

Jan 2025 — Jul 2025

Work

Production Operator — Spot Welding (Vocational Program)

PT Toyota Motor Manufacturing Indonesia (TMMIN)

- Operated spot welding machines on the production line according to SOPs, output targets and quality standards
- Applied 5S and workplace safety (K3) in the work area

Sep 2022 — Sep 2023

Work

Mechanic (Industrial Internship)

PT Bahari Prima Manunggal

- Troubleshot the K800FT bandsaw machine and replaced damaged components
- Assembled and installed new control box panels for the Baader 600 fish fillet machine and the Bitzer refrigeration unit

2020 — 2024

Education

Mechatronics Engineering

SMKYPPT Garut

- School project — MPS Sorting: wiring, pneumatics and Omron PLC programming

Certifications & training

BNSP

Welding Spot — **2026**

Let's work together.

Open to job opportunities and automation projects, from PLC programming and control panels to machine wiring. Reach me by email or LinkedIn.

Project videos and the interactive version of this portfolio are at **littlegusz.github.io**

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